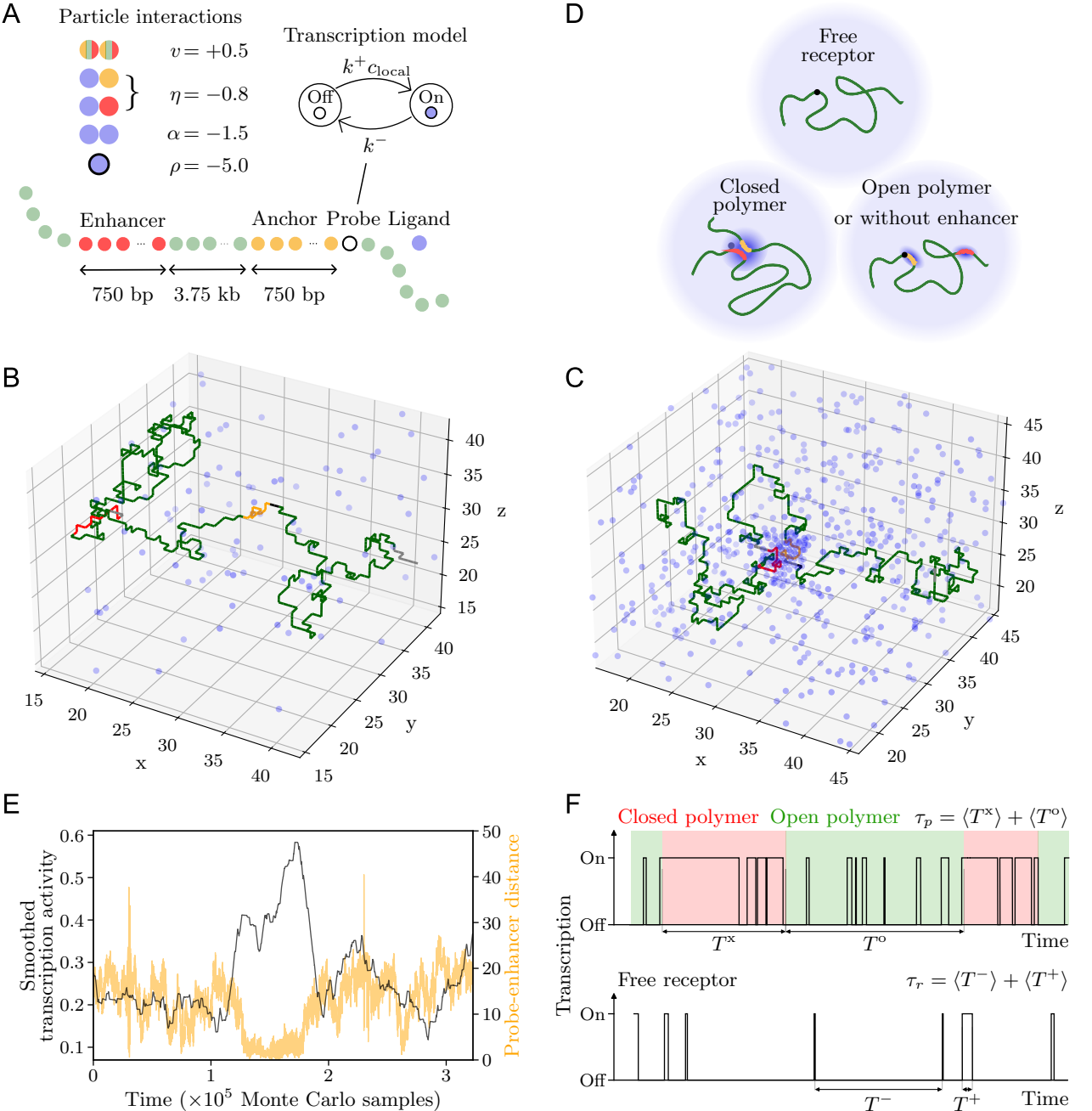

A
Particle interactions
$v = +0.5$
$\eta = -0.8$
$\alpha = -1.5$
$\rho = -5.0$
Transcription model
$k^+ c_{\text{local}}$
Off
On
k^-
Enhancer
Anchor
Probe
Ligand
750 bp
3.75 kb
750 bp
D
Free receptor
Closed polymer
Open polymer or without enhancer
B
x
y
z
C
x
y
z
E
Smoothed transcription activity
Probe-enhancer distance
Time ($\times 10^5$ Monte Carlo samples)
F
Closed polymer
Open polymer
$\tau_p = \langle T^{\text{x}} \rangle + \langle T^{\text{o}} \rangle$
Transcription
On
Off
T^{x}
T^{o}
Time
Free receptor
$\tau_r = \langle T^- \rangle + \langle T^+ \rangle$
On
Off
T^-
T^+
Time